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FULL ARTICLE

Spatial patterns of Cultural and Creative Industries: Creativity and *filière* behind concentration

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Abstract

Spatial concentration of Cultural and Creative Industries (CCIs) is not a new topic in academic research. However, the analysis of this phenomenon often neglects that CCIs behave differently, due to their heterogeneity. Building on the literature on CCIs' concentration, a novel classification of CCIs is presented based on two key dimensions: heterogeneous creativity and *filière*. CCIs spatial concentration is tested along these two dimensions, highlighting that the location determinants differ at their intersection. Results renew the interest on CCIs' clustering and define new perspectives for policy interventions.

KEYWORDS
creative value chain, cultural and creative industries, innovation, localization economies, urbanizationc economies

JEL CLASSIFICATION
R10, R12

1 | INTRODUCTION AND AIM OF THE RESEARCH

Academic research on Cultural and Creative Industries' (CCIs') location is not new in economics. Right from the beginning, CCIs showed a peculiar, concentrated spatial distribution, with only a few areas offering the conditions for their concentration (Namyślak & Spallek, 2021) and in three out of four cases these areas are metropolitan ones (Boix et al., 2015).

There are many examples of CCIs clustering, areas where the condition for the flourishing and the concentration of CCIs are favourable. Local specificities forge the creative expressions and stimulate the emergence of creative ideas and innovations. As examples of places where CCIs cluster, the Brera Design District in the centre of Milan is proof of how the constant engagement of local stakeholders helps businesses flourish, supporting the emergence of a local identity.

In Berlin, the northern district of Prenzlauer Berg is a melting pot of styles and people coming from different areas. Trendy cafés, pubs, clubs, restaurants, galleries and theatres offer a proper cultural and creative atmosphere for the generation of creative ideas.

Moreover, with over 30 art galleries and studios, the small Provençal village of Mougins aims at maintaining its artistic heritage forged by Picasso and other creators.[1]

Culture and creativity are at the basis of CCIs' concentration: culture is deemed as the glue of postmodern geography (van Oort, 2008) and creativity flourishes locally, taking advantage from the system of local interconnections (Comunian et al., 2010). CCIs proved to be more agglomerated than other businesses, especially at short distances (Coll-Martínez et al., 2019).

However, despite a general trend, reasons for CCIs' clustering are industry-specific (Comunian & England, 2019) depending either on the structure of the sector (e.g., cultural heritage-related industries cluster where cultural heritage is located) or on the system of interactions with consumers and other businesses (e.g., cinemas cluster where consumers are).

This work proposes a novel comprehensive conceptual perspective based on the following aspects. First, CCIs are composed by an extremely diversified set of industries, whose *creative* activity is heterogeneous and it is mirrored by the different forms of innovative output they may generate. Second, territories shape the *creative* capacity of CCIs resulting in different intensity levels across regions.[2] Finally, CCI's position within their *filière*, that is, the trade relationships they engage, together with different creativity levels is capable of explaining CCIs' spatial patterns.

Therefore, the factors driving CCIs clustering at the local level, also called location determinants, are identified at the intersection of creativity and *filière*.

With this aim, the work is structured as follows. First, Section 2 classifies the existing literature on spatial concentration of CCIs, revealing that creativity and trade relationships represent the two main logics behind spatial clustering (subsection 2.1). Moreover, based on the literature discussion, the new research trajectories are presented. The two dimensions are defined and used to classify CCIs accordingly. Then, a conceptual taxonomy of the main prevailing determinants of CCIs' spatial concentration based on creativity and *filière* is presented (subsection 2.2).

Section 3 discusses the issue of measurement of creativity and *filière* in this work. Section 4 outlines the empirical model and discusses the results obtained together with a set of robustness checks. Finally, Section 5 concludes with comments and future perspectives.

2 | LOCATION DETERMINANTS OF CCIs: STATE OF THE ART AND NEW RESEARCH TRAJECTORIES

2.1 | State of the art

Defined in many ways, such as those industries “which have their origin in individual creativity, skill and talent and which have a potential for wealth and job creation through the generation and exploitation of intellectual

[1] https://www.theguardian.com/travel/2011/jul/24/france-mougins-art-picasso. Accessed on 20 September 2022.

[2] In fact, the context provides inputs for creativity in a collective learning process through exchanges and cross-fertilization (Camagni, 1991; Camagni & Capello, 2002).

property" (DCMS, 2001),[3] CCIs have always proven a distinguished innovative capacity derived from their creative behaviour.

Among their features, CCIs' spatial pattern soon found slarge space in the academic debate (Caves, 2006; De Vaan et al., 2013), also because the tie between creativity and places has ancient roots.[4] CCIs' clustering was found to be heterogeneous, described mostly as an urban phenomenon (Power, 2011), due to the presence of specific amenities attracting creatives (Florida, 2002), but also typical of historical non-urban cultural districts (Bertacchini & Borrione, 2013; Santagata, 2002), where local culture and traditions are handed down from generation to generation.

The reasons behind this interesting localization pattern are several and potentially overlapping, and confusion still exists. This review presents four conceptual approaches, aimed at classifying the determinants of CCIs' spatial clustering under two conceptual dimensions: the heterogeneous creativity of CCIs and their industrial relationships (*filière*).

On the one hand, the heterogeneous creative capacity of CCIs mirrors the difference between static and dynamic gains. Static benefits refer to the efficiency gains due to reduction of transaction and production costs and general productivity increase for firms. Dynamic benefits, instead, refer to the innovative gains made possible by easiness of information exchanges, face-to-face interactions, cultural homogeneity and mutual understanding.[5]

On the other hand, the industrial relationships through trade exchanges embed conditions behind localization and urbanization economies.

2.1.1 | Industrial district approach

The *Industrial District* approach, whose name comes from the nature of the place where benefits of agglomeration arise (Becattini, 1989), describes the determinants of the spatial concentration of actors especially within the same sector due to localization economies, that is, when firms in the same sector locate close to each other. Many are the historical examples of industrial districts, especially of CCIs, such the jewellery sector in Birmingham, and the textile in Prato. Santagata (2002) referred to them as *Cultural Districts*, where the industrial specialization in cultural activities meets a larger cultural identity and where people add economic value through their creativity.[6]

District economies mirror the combination of various endogenous components, shaping "the overall scale of activity in an area, thereby affecting the productivity of all firms" (Capello, 2002, p. 388), and they are associated to the (un)relatedness of the employment structure (Lazzeretti et al., 2012) and to the consequent capacity of places to create value around the district (Boix & Trullén, 2010), also through place branding (Jansson & Power, 2010). The spatial concentration of firms within the same sector is associated to *labour pooling* (Duranton & Puga, 2004), that is, the development of a specialized labour force through the cumulation of sector-specific skills. This process is relevant for manufacturing CCIs as it requires specific knowledge for the production, resulting from ancient traditions handed down from generation to generation within districts (Santagata, 2009).

In addition, the specialization of an area triggers the rise of activities, either on top or at the bottom of the production chain, capable of valorizing the local production in a process known as *input sharing*. Considering CCIs, the role of the *filière* in explaining their clustering has been weakly touched upon by the literature (see Boix, 2013 analysing publishing and printing), although its importance is acknowledged (Santagata, 2009). This approach highlights

[3]Cfr. Hartley (2021) for an advanced introduction to creative industries and the multiple definitions proposed by the literature.

[4]History is full of examples of creative places where the creative genius flourishes (cfr. Andersson, 2011 for a critical discussion of the local conditions behind creative places).

[5]The study of the benefits from agglomerated settings for firms dates back in the literature, building, among others, on the masterpieces by Hoover (1937), Marshall (1920) and Ohlin (1935). The importance to distinguish between static and dynamic agglomeration economies is also due to Porter, who distinguished the elements behind fruitful learning for innovation (Porter, 1990), that is,. the dynamic ones, from the traditional efficiency gains, often called static gains. The places in which different benefits arise depend on the characteristics of each territory, capable to sustain the process of innovation and transformation (Camagni et al., 2016) with important repercussions for economic growth (Garcia-López & Muñiz, 2013).

[6]For a review of the concept of Cultural and Creative Clusters, cfr. Chapain and Sagot-Duvauroux (2020).

that efficiency gains for CCIs emerge thanks to industrial relationships with similar and related industries, due to static localization economies.

2.1.2 | Cultural environment approach

CCIs cluster also (and especially) in cities to exploit the so-called urbanization economies that stimulate firms' performances (Ho & Sheng, 2022). Large metro areas allow CCIs to reduce the gap with institutions, whose support matters for the development of creative projects (Christopherson & van Jaarsveld, 2005). The presence of a supportive institutional environment stimulates the competitive advantage in the form of increasing returns to scale and scope, functioning as “a seedbed of creativity and innovation for the industry” (Scott, 2002, p. 965).

Moreover, the nature of larger markets, typical of urbanized areas, is beneficial for CCIs because of the proximity to a large set of consumers (Yusuf & Nabeshima, 2005), reducing transportation costs and deepening the profitability also for early-stage firms (Campbell-Kelly et al., 2010). Moreover, cities enhance *geographical* economies of scope, that is, a close geographical collocation of related capabilities triggers efficiency in production (Florida et al., 2012), allowing cultural industries to benefit from the synergies generated by close proximity to related skills, inputs and capabilities.

Finally, as urban areas host high-level functions, CCIs benefit from an easier access to them. Industrial diversity supports the creation and promotion of cultural contents, like the case of financial and legal services for the valorization of Intellectual Property Rights (IPRs). Therefore, cities incubate diversified firms and workers, allowing for an easy allocation of creative workers and an easy match between firms and support activities (Lazzeretti et al., 2012; Sánchez Serra, 2016; Tao et al., 2019). Conversely with the previous approach, higher efficiency is generated thanks to industrial relationships with unrelated actors in a vast economic environment, due to static urbanization economies.

2.1.3 | Cognitive proximity approach

In a context of efficiency gains, the first two approaches highlighted the difference between localization and urbanization economies. However, CCIs' clustering responds also to a dynamic stimulus, that is, specific areas (urban or not) stimulate their innovative capacity, reducing the dynamic uncertainty and the business risk, and increasing entrepreneurial creativity and innovation (Camagni, 1993). Literature proposed different labels to describe the local conditions triggering firms' innovativeness, like *innovative milieu* (Camagni, 1995), *learning region* (Asheim, 1996) and also *pôles de compétitivité*.[7]

Here, the interest is to identify the determinants that stimulate the flow of knowledge and the emergence of creative ideas (Potts, 2019), namely the *dynamic* agglomerative benefits.

CCIs benefit from the knowledge flow acquired through a process of learning based on *making* and on *interacting* (Santagata, 2009). This flow of knowledge, especially tacit, is fostered within creative clusters of similar industries like the districts where the exchanges are easier and when actors understand each other (Capello, 2009) thanks to *cognitive proximity*.[8] The related variety approach (Boschma & Iammarino, 2009; Frenken et al., 2007) was applied to CCIs to explain the process of cross-fertilization and cognitive bonds among different industries. Related variety is expected to trigger creativity and innovation in CCIs (Innocenti & Lazzeretti, 2019; Klement & Strambach, 2019); especially, also enhancing clustering at the local level (Lazzeretti et al., 2012). The success of CCIs cluster is due exactly to tacit knowledge diffusion (O'Connor, 2009), transforming the cluster into a “creative field” (Scott, 2006).[9]

[7]https://www.entreprises.gouv.fr/fr/innovation/poles-de-competitivite/presentation-des-poles-de-competitivite.
[8]The label cognitive proximity builds on Nooteboom (2000).
[9]Literature considers other forms of proximity like organizational and institutional ones in explaining the sources of knowledge creation and diffusion (Knoben & Oerlemans, 2006; Torre, 2014). Despite their relevance, they are more associated to industrial organization and management studies.

In this approach, dynamic localization economies emerge thanks to industrial relationships with similar industries. Due to cross-fertilization of ideas from related domains, CCIs show a stronger innovative performance.

2.1.4 | Competitive market approach

There are some other factors behind the innovativeness of CCIs when nestled within vast and diversified (usually urban) environments. Competition effects stimulate the innovativeness of creative actors that innovate for distinguishing themselves with respect to the rest of the market. What matters is the ability to differentiate from competitors and this happens when the variety of inputs available within the market is high and heterogeneous. Therefore, CCIs' creativity benefits from dense urbanized environments for many reasons. Cities allow the exploitation of the economies of variety, that is the set of benefits that result from variation, meaning that trying a variety of things improves the creative capacity distributed throughout the system (Rao, 2016). Plus, a varied and stimulating environment, where the economic and personal relationships generate ideas, feeds CCIs (Currid, 2007). Compared to the cognitive approach, CCIs may also benefit from unrelated domains. There is not only a need of understanding, but new disruptive innovations may come from the application of ideas generated in other fields. Finally, it is important to stress the role of consumption in explaining the forces behind spatial clustering (Glaeser et al., 2001). In fact, CCIs' products do not respond to a classical logic,[10] but they refer to the field of experience of consumers that drive their innovative performance. In this last approach, dynamic urbanization economies have the form of diversified industrial relationships and even interactions with consumers, able to explain higher creativity of CCIs.

2.2 | New research trajectories

This review of the literature on spatial clustering of CCIs highlighted two main dimensions of interest to interpret the phenomenon: the industrial relationships and the creativity embedded in the innovative performance.

Considering the former, industrial relationships are at the basis of the concept of *filière*. Although the word *filière* can have different meanings based on the field of application, it is generally defined as a sequence of conception, R&D, sourcing, production and distribution phases (Bianchi & Labory, 2013). It reflects both the steps required for product generation (i.e., transactions among firms) and the final market sales (i.e., consumers' purchases).[11] According to this logic, the *filière* accounts for market transactions of firms and industries with other segments of the economy, either other businesses or the consumers (Montaigne & Coelho, 2012; Raikes et al., 2000). Thus, to understand *filières* a discussion of the industrial co-operation considering both similar and different industries is worthwhile (Chesnel et al., 2013).

In order to capture CCIs' industrial relationships, this work identifies three *filières* each CCI may belong to:

1. *Creative filière:* if the main trade partner is represented by other CCIs;
2. *Vast filière:* if the main trade partner is represented by industries not belonging to CCIs; and
3. *Short filière:* if the main trade partner is represented by final consumers.

As an example, consider three industries belonging to CCIs: printing, computer programming and retail of fashion goods. Printing activities belong to the creative *filière*, as their main trade partner is expected to be a publishing sector (i.e., another CCI). Computer programming, instead, is part of the vast *filière*, as its services are offered to any

[10]CCIs' products do not follow a classical economic logic because they embrace a specific meaning for people who consume them, stressing their socio-cultural role. Being in between arts and commerce, their essence can be retrieved in the experience offered to the consumers. In this sense, interaction with consumers is so relevant because the experience offered and the meaning embedded represent the true innovation.
[11]Some considered *filière* as the French equivalent to the concepts of value chain and supply chain (Lançon et al., 2017).

other industry of the economy, not mainly to CCIs. Eventually, retail activities fit the short *filière*, due to their natural market orientation. Hence, in this work, the *filière* captures the prevailing partner of CCIs' trade exchanges.

Although the *filière* structure is an industrial dimension, when translated at the local level it reveals the importance of trade interactions within an area as they allow actors to better co-operate and especially to mitigate the risks of the production of cultural products through both territorial specialization and product quality increase (Chesnel, 2015).

The concept of *filière* is useful for studying CCIs' clustering because it helps explain different agglomerative benefits, mirroring localization and urbanization economies, as discussed in the review. Different *filières*, in fact, reflect localization or urbanization advantages. Localization economies are based on proximity to related actors; urbanization economies, instead, on proximity to unrelated ones.

The next section will discuss how each CCI has been classified in one of the three *filières*, based on data.

Considering the latter, instead, some discussion on the forms of creativity is needed. The complexity in defining[12] creativity itself (Runco & Jaeger, 2012) affects also CCIs-related issues. This work adopts an industrial perspective on creativity, with a focus on the output it generates and therefore on the forms and intensity of CCIs' innovation.[13] Although CCIs are considered among the most innovative actors of the economy (Müller et al., 2009), literature generally overlooks that they innovate in different forms and at different intensities and especially that this depends on the territory where they locate. In fact, they are not equally creative both in the forms (generating various forms of innovations, ranging from patents to copyrights) and in the intensity (either creating something new or adopting other innovations, through replication). CCI's innovative behaviour is important for concentration because they respond to different static (efficiency) and dynamic (innovative) agglomerative benefits and the heterogeneous creativity mirrors this difference.

The original perspective on CCIs' creativity of this work is based on two aspects. First, in order to account for the heterogeneous forms of creativity in CCIs, three *creative modes* can be identified: *technological*, *symbolic* and *artistic* creativity. Technological creativity reflects scientific and technological ideas, generally industrially applicable like for hardware and software industries. However, technological ideas are not only typical of high-tech sectors but they can be associated to others (e.g., patenting in the hospitality sector; Succurro & Boffa, 2018).

Symbolic creativity mirrors the symbols and the images that firms show on the market, capturing the economic value at the industrial level, like trademarks in the retail sector. Trademarks are nothing but symbols used to protect the business-identity and products, of any kind.

Artistic creativity captures imagination and the ways to describe the world through texts, sounds and images, as in the publishing and printing industries. Artistic creativity is not only typical of art-related businesses but it may refer to support activities that contribute to the generation of the piece of art.

Second, the industry-specific capacity of generating an innovative output depends on the place where the industry is located. In fact, based on the territory where CCIs locate, the territorial environment shapes the innovative capacity of these actors, attracting different functions and different talents. In this sense, territories act as incubator of new ideas and attractor of creative talents (Glaeser & Maré, 2001) with firms internalizing most of the territorial features (Paniccia et al., 2015). These processes, often known as *collective learning*, have been investigated by the regional economics literature (Camagni, 1991; Camagni & Capello, 2002; Lawson & Lorenz, 1999) that discussed the contexts' role in providing inputs for innovation. Considering CCIs, local culture fuels *creativity* of economic actors (Santagata, 2009), making it a pivotal factor of the territorial capital of places (Camagni, 2009) capable of triggering local innovation.

[12]The literature extensively discussed the potential interpretations of creativity, considering it as a way of dynamically restructuring the system of information, starting from an existing knowledge environment to fashion new knowledge and competence (Andersson, 1985). However, no agreement exists whether it is reflected in the process, the person or the product (Simonton, 2011).

[13]Literature also studied the topic through a people-related approach. According to this psychological dimension, the focus is on the creative individual generating creative ideas and on their determinants such as personality, openness, independence, but also lower tendency towards psychopathology (Simonton, 2009). This segment of literature was enlarged by the economics literature on the creative class (Cerisola, 2018; Florida, 2002; Florida & Mellander, 2020).

The classification of creativity in different forms is rooted in the vaster theory of *differentiated knowledge bases* because innovation differs substantially across industrial domains and different activities rely on different knowledge bases (Asheim, 2007; Asheim & Gertler, 2006). However, generally CCIs are associated mostly, if not only, to one specific knowledge base (e.g., the symbolic one in Asheim's work) not considering that the knowledge bases may differ also within CCIs themselves. In this work, instead, the three forms of creativity reflect the fact that the CCIs are grounded in different *creative bases*.

The new framework presented in this work accounts for this complexity, considering that CCIs innovate in different ways, considering various innovative forms, including aesthetics, functionality and content (Stoneman, 2010), and that creative intensity depends on territories where CCIs locate, distinguishing *Inventive* from *Replicative* activities.

Therefore, a region-industry framework is described using Figure 1. Within each region, each industry belonging to CCIs is evaluated according to its performance in different creative modes, assigned either to the group of *Inventive* or to the *Replicative* one based on this performance levels.

Thus, if *filières* and creativity in CCIs mirror different agglomerative benefits, the determinants of their spatial pattern should be studied at their intersection. A taxonomy of typologies of CCIs based on different types of creativity and *filière* is proposed here (Figure 2), to highlight for each type of CCI the prevailing determinants of spatial concentration. In the rows, three *filières* are presented, while in the columns CCIs are subdivided according to their creative intensity.

Considering activities under the *Creative filière*, sectoral and cognitive relatedness are compared. The former refers to the presence within the districts of sectoral-related employees. This variable should influence the localization choices of less innovative CCIs in a labour market logic because activities with creative *filière* that do not intensively innovate reduce the cost of matching the related labour supply within the district. Cognitive relatedness,

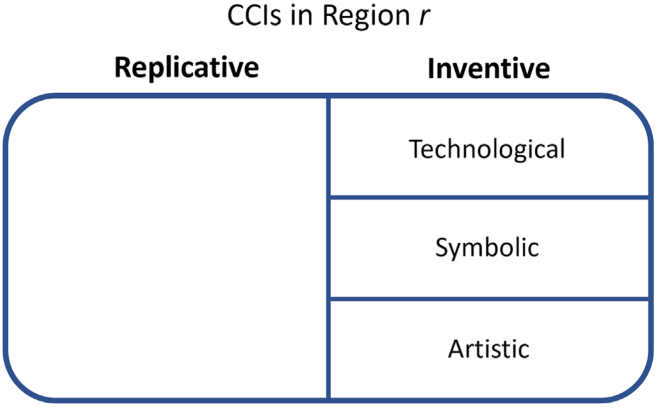

FIGURE 1 Regional framework for CCIs classification based on creativity.

Creativity / ***Filière***	**Replicative**	**Inventive**
Creative *filière*	Sectoral relatedness	Cognitive relatedness
Vast *filière*	Sectoral unrelatedness	Cognitive unrelatedness
Short *filière*	Labour pooling / *Market size*	Creative knowledge accumul. / *Tech. in retail*

FIGURE 2 Prevailing determinants of CCIs concentration by creativity and *filière*.

instead, refers to presence of sectors that are related due to complementary competences in a cognitive based definition that stimulates communication and interactive learning among different industries. This should promote the innovativeness under a more dynamic logic (Frenken et al., 2007). This is a *collective learning* variable as it considers the cognitive knowledge base as source of relatedness. People learn and innovate more easily in a domain if they are already familiar to it. Hence, the expectations for the first row are that both forms of relatedness are positively associated to spatial concentration of actors but for distinct groups, according to the heterogeneous innovative capacity.

Second, the *Vast filière* is the domain of unrelatedness. The sectoral unrelatedness measures how diversified is the sectoral composition of CCIs within the area. Clustering of *Replicative* activities with a vast *filière* should benefit from it as the diversity of the labour force represents a factor of attraction to be closer to support activities. Cognitive unrelatedness, instead, refers to presence of a heterogeneous cognitive environment due to the presence of creative knowledge from distinct groups of CCIs. Although unrelated variety is often considered a negative driver for productivity and innovation of firms (Aarstad et al., 2016), the cognitive unrelatedness should drive spatial concentration of innovative CCIs if they rely on a vast *filière*. Vast and heterogeneous environments may provide a positive stimulus to innovation, due to cross-fertilization from other sectors. In pair with the previous one, cognitive unrelatedness also refers to *collective learning*.

Finally, the bottom row is related to activities that concentrate most of their trade relationships with final consumers, both private households and public entities. At this point, it is necessary to distinguish between service- and manufacturing-related activities. Within the *Short filière*, manufacturing and services activities as they participate to the value chain in different ways. Considering manufacturing, *Replicative* CCIs will look for a favourable labour market. That is why labour pooling—representing the availability of professionals within the district—helps reducing costs due to specific skills and knowledge present at the local level.[14] Instead, *Inventive* CCIs are expected to concentrate in places where creative knowledge is rich and widespread, due to cumulative processes of knowledge accumulation and learning (Caragliu & Nijkamp, 2012; Cohen & Levinthal, 1990). Considering services, the main driver of spatial concentration of *Replicative* CCIs with a short *filière* should be the market potential while for *Inventive* ones consumers stimulate innovation in this segment with their growing propensity to novelties (Pantano, 2014; Pantano & Di Pietro, 2012). Hence, the concept of technology in retail as a trigger to spatial concentration of innovative service CCIs in this segment is included.

Summing up, the main determinants of CCIs' clustering are presented, stressing on the importance of heterogeneous creativity and *filière* as main conceptual approaches. Hence, a conceptual taxonomy of these determinants is proposed, intersecting creativity and *filière* in CCIs. The determinants identified differ along the two dimensions, describing the phenomenon of clustering in an innovative, more comprehensive way. Once the novel framework is presented, the aim of this work is to empirically test the taxonomy, highlighting that CCIs are heterogeneous, and their behaviour should be studied accordingly. Hence, the next section discusses the issue of measurement of CCIs' features presented in the taxonomy. Then, the empirical model is presented and tested in Section 4.

3 | MEASUREMENT OF CREATIVITY AND *FILIÈRE* IN CCIs

3.1 | Creativity in CCIs

In any CCI-related work, setting the industrial boundaries is the first issue. Here, the list of industries presented by Santagata (2009) is chosen for two main reasons. First, it assumes that creativity may express differently in different sectors, as claimed in this work. Second, the method is extremely accurate due to the fine sectorial disaggregation.

To the CCIs list by Santagata (2009), the reasoning presented in Figure 2 is applied, empirically separating *Inventive* from *Replicative* activities. Due to intangibility of creativity (Cerisola, 2019), the innovative outputs of CCIs are

[14]Labour pooling has mostly been associated to manufacturing industries in the empirical literature (Overman & Puga, 2010; Rigby & Brown, 2015).

used as a proxy of creativity. To capture all forms of creativity and their expressions, patent, trademark and copyright intensity are used to proxy the creative modes. Using IPRs as a proxy of creativity means considering innovation as the output of the creative process (Wohl, 2021).

The creative performance of a sector in a region is captured through a dummy variable that takes value 1 if:

- the number of IPRs (either trademarks or patents), per employees is higher than the overall EU level, as in (1);

$$\frac{IP_{i,r}/E_{i,r}}{IP_{i,EU}/E_{i,EU}} > 1, \tag{1}$$

where *IP* refers to Intellectual Property (either patent or trademark), *E* is the employment and *i* and *r* are, respectively, a generic sector and a generic region.

- the sector is copyright intense according to the list provided by EPO and EUIPO (2016), used as reference.[15]

In this way, each CCI in each region is assigned to the *Inventive* (*Replicative*) group, according to its higher (lower) innovative capacity in one of the possible creative output.

The binary approach to creativity is used for three main reasons. First, even if it is empirical, this approach allows to classify CCIs and to assign dummy variables to each industry in each region to one of the possible groups with no ambiguity. Second, the overlapping of *Inventive* and *Replicative* could create confusion and could make results more complex to be read. Finally, due to the lack of data for copyrights, the binary approach was forced for this creative mode. Hence, this is applied to all forms of innovation, in order to make this classification germane to different contexts.

In this framework, the choice of both industrial and geographical scale is important. Considering the former, this work translates the classification made by Santagata (2009) for Italian sectors into a European one, made of NACE four-digit sectors (classes). This level of disaggregation is accurate, and it allows us to select only activities that have a true cultural/creative essence and measure the phenomenon of clustering precisely (the detailed list of sectors is presented in the Appendix).

Considering the latter instead, the scope of this analysis is the entirety of NUTS 3[16] regions of EU 28 (including UK)[17] in 2015. NUTS 3 regions represent small spatial entities (max. 800,000 average inhabitants),[18] allowing to properly capture CCIs' clustering in space.

CCIs' employment at the local level is measured using Orbis[19] data, to measure trademarks, patents and employees with a fine industrial and geographical disaggregation, ensuring the comparability across places.[20]

3.2 | *Filière* structure of CCIs

The second dimension of interest is the *filière* structure of CCIs. In order to describe industrial linkages, input–output tables (IOTs) from FIGARO experimental tables (European Commission, 2018) were considered, mapping industrial relationships. With a specific focus on CCIs, from the industry-by-industry table, the linkages of each industry with all others are outlined, explaining the transactions of intermediate consumption and final uses.

[15] Since copyrights are authors' and not firms' rights, they are difficult to be localized in space.
[16] NUTS is the acronym for Nomenclature of Territorial Units for Statistics from the French Nomenclature des Unités Territoriales Statistiques.
[17] Data for Northern Ireland are excluded due to inconsistencies found in Orbis.
[18] European Commission (2003).
[19] The Orbis database, product of Bureau van Dijk, is the world platform for exploring data on companies.
[20] Orbis collects the amount of IPRs produced by firms, grouped at the region-industry level.

CCIs are classified into the three *filières* considering the trade flows coming from the IOTs and distinguishing between intermediate and final-demand flows. More specifically, if the share of trade flow towards the final demand over the total flow exceeds the mean of all CCIs, the industry is assigned to the "Short *filière*."

Then, considering the remaining industries, if the share of intermediate trade flows with other CCIs over the total intermediate flow exceeds the mean of all CCIs, the sector is assigned to the "Creative *filière*," otherwise to the "Vast *filière*."[21] Table A2 in the Appendix presents CCIs' classification into different *filières*.

3.3 | CCIs clustering and its determinants

As presented in the theoretical discussion, CCIs' concentration is more complex compared to what the traditional literature claims. CCIs' creativity, as measured in subsection 3.1, and *filière*, as measured in subsection 3.2, influence the spatial pattern, requiring a deeper analysis of the prevailing agglomeration factors.

This work proposes a conceptual taxonomy of CCIs' prevailing determinants behind spatial concentration, claiming that at the intersection of creativity and *filière* they differ. In this regard, the measurement of both CCI's spatial concentration and the variables included in the taxonomy are required.

Concerning CCIs' concentration, matching Orbis information with regional employment from Eurostat,[22] the regional employment share for each CCI is computed as in (2) and used as dependent variable for the analysis:

$$share_{i,r} = \frac{empl_{i,r}}{empl_r}. \tag{2}$$

This variable captures CCIs clustering and an overview of their concentrated behaviour is presented through the descriptive statistics in Appendix Table A1. Although all sectors display high asymmetry in the distribution (i.e., skewness), some sectors are more concentrated than others. Specialized activities such as design, publishing and motion pictures are more concentrated than manufacturing of food, radio and television broadcasting that are more evenly spread across European regions. Also within arts, entertainment and recreation activities, CCIs are heterogeneously clustered. Notable is the difference between performing arts (9001) and artistic creation (9003), the former is much more geographically widespread than the latter. Production of theatrical presentations, concerts or opera is typical of many places, while the pure creation of new pieces of art (e.g., activities of sculptors, painters, writers) takes place in specific areas, where artists find the conditions to cluster. As this work takes a general perspective on the concentration of all CCIs, industry-specific fixed effects are therefore required.

Moving towards the determinants of clustering, relatedness and unrelatedness measures associated to the first two rows of the taxonomy can be calculated using related and unrelated variety measures, respectively. Unrelated variety captures the entropy of the two-digit industrial distribution (Equation 3); related variety is the weighted sum of the entropy at the four-digit level within each two-digit class (Equation 3) (Frenken et al., 2007)[23]:

$$UV = \sum_{g \in G} P_g log_2 \left(\frac{1}{P_g}\right), \tag{3}$$

$$RV = \sum_{g \in G} H_g P_g. \tag{4}$$

[21]If the logic is replicated using the median and not the mean, minor changes apply. In this way, the three *filières* are mutually exclusive.

[22]Besides Eurostat, robustness checks have been performed using ARDECO data for regional employment. No major changes emerge.

[23]P_g is the share of the 2two-digit sector over the total economy, obtained as the sum of the 4four-digit shares belonging to it: $P_g = \sum_{i \in S_g} p_i$ and G the set of all two-digits codes. Instead, the weight is $H_g = \sum_{i \in S_g} \frac{p_i}{P_g} \log_2 \left(\frac{P_g}{p_i}\right)$.

Relatedness and unrelatedness among economic activities can be calculated using different variables, influencing the interpretation. Referring to the taxonomy, the indicators of this work are calculated in two diverse ways. First, both RV and UV on the left-hand side of the table, related to *Replicative* CCIs, are computed through a sectoral entropy, using the employment levels in CCIs at the two different scales. On the other hand, the two *cognitive* indicators of the taxonomy are obtained using patents. Thus, P_g and p_i are calculated using employees and patents for sectoral and cognitive (un)relatedness, respectively. In the former case, relatedness refers to the workforce in general, describing the degree of sectoral (un)relatedness in CCIs. In the latter case, instead, the indicators refer to the concept of *cognitive proximity*.[24] The choice of patents instead of other forms of innovation builds on two reasons: the granularity of data and the nature of patenting itself. Patents (available at a fine scale) mix knowledge from both similar and also different fields, and it is possible to say that they intrinsically embed knowledge.

Indicators used in the bottom row of the taxonomy differ considering manufacturing and services separately. The spatial concentration the former is tested against variables in (5) and (6), representing labour pooling (LP) and creative knowledge (CK), respectively:

$$LP_{i,r} = \frac{\sum_{est \in i} |\,\%\Delta E_{est(i),r} - \%\Delta E_{i,r}\,|}{n}, \tag{5}$$

$$CK_{i,r} = \frac{patents_{i,r}}{patents_r}. \tag{6}$$

Building on the indicator by Krugman (1991), the variable in (5) was proposed by Overman and Puga (2010) and re-adapted by de Almeida and de Moraes Rocha (2018). It captures the pooling effect measuring the difference (in absolute value) between the growth rate of employment of an establishment and the same growth rate in the industry *i* which the establishment belongs to. If establishments and industries adapt in different ways to labour shocks, the value is larger. This being an establishment-specific indicator, the industry-specific value is obtained averaging all establishments for each industry. Instead, the measure for the local creative knowledge accumulated is constructed through the sector-specific share of patents over the total number of patents. In this way, it measures the intensity of creative knowledge for each sector locally.

Focusing on services, the market size is proxied through the local value added. It is a suitable indicator for the market potential because high value added generated can be associated to large potential consumption of final goods. Finally, the population purchasing holiday services online (online sales) is deemed as a good proxy for the regional pervasiveness of technology in the retail market. However, online sales are strongly correlated with the education levels and economic size of the region, meaning that places where both the education level and the size of the economy is high are also places where people buy substantially online, in line with research on the determinants of e-shopping (Naseri & Elliott, 2011). These determinants may confound the effect of technology adoption on the higher innovativeness of market-oriented CCIs. In the attempt to fix this issue, residuals of a first-stage regression of online sales against education levels and regional economic size are used as explanatory variable.[25]

[24]As a technical note, indicators of related and unrelated variety are computed only within the *Macrosector* of CCIs for two reasons. First, in this way, the measures are creative in nature, indicating the sectoral and cognitive (dis)order of the regional creative system, with a conceptual impact on the interpretation. Second, for the sectoral measures a so wide and detailed dataset was not available. For symmetry, the same methodology was applied for both sectoral and cognitive measures.

[25]Though this procedure does not find unanimous support in the literature (Freckleton, 2002), residuals of a first-stage regression are often employed to control for spurious effects in multivariate setting.

4 | EMPIRICAL MODEL

4.1 | Empirical setting

With the aim of appraising whether the intensity of the agglomeration of *Inventive* and *Replicative* CCIs within each *filière* is affected by different elements, the following model is employed:

$$share_{i,r} = \begin{cases} f(SectRel_r; CogRel_r; Inv_{i,r}; X) + \varphi_i + \mu_c + \varepsilon_{i,r} & \text{if } i \in Creative\,fil. \\ f(SectUnrel_r; CogUnrel_r; Inv_{i,r}; X) + \varphi_i + \mu_c + \varepsilon_{i,r} & \text{if } i \in Vast\,fil. \\ f(CK_{i,r}; LP_{i,r}; Inv_{i,r}; X) + \varphi_i + \mu_c + \varepsilon_{i,r} & \text{if } i \in Short\,fil. \& i \in Manuf. \\ f(MktSize_r; Tech.retail_r; Inv_{i,r}; X) + \varphi_i + \mu_c + \varepsilon_{i,r} & \text{if } i \in Short\,fil. \& i \in Serv. \end{cases} \tag{7}$$

To achieve such an aim, both a baseline regression and an augmented one have been estimated for each equation in (7). The baseline regression tests both variables of each row alone while the augmented one interacts the *Inventive* dummy, as identified in subsection 3.1, with the variables of interest. In this way, it is possible to assess the differentiated determinants explaining the spatial concentration of *Inventive* and *Replicative* CCIs. To clarify, taking the first equation as an example, the regressions estimated displays the following functional form:

$$\begin{gathered} share_{i,r} = \beta_1 SectRel_r + \beta_2 CogRel_r + X\boldsymbol{\gamma} + \varphi_i + \mu_c + \varepsilon_{i,r} \\ share_{i,r} = \beta_1 SectRel_r + \beta_2 CogRel_r + Inv_{i,r} * (\beta_3 + \beta_4 SectRel_r + \beta_5 CogRel_r) + X\boldsymbol{\gamma} + \varphi_i + \mu_c + \varepsilon_{i,r} \end{gathered} \tag{8}$$

The model is an industry-region framework and it is estimable through a panel regression. In this case, a Random Effect framework is employed as a fixed effects would not support variables that are constant in one of the two dimensions. However, all equations control for two-digit industry (φ_i) and country (μ_c) fixed effects.

Finally, in order to reinforce the analysis, a set of control variables X aligned with the literature is included. They range from population density, capital city region (dummy), number of UNESCO Sites, total population, new EU member after 2004 (dummy), GDP *per capita*, education (Population 25–64 with tertiary education). Moreover, the explanatory variables used have been lagged (if possible) to cope with potential endogeneity coming from simultaneity. Table 1 presents the descriptive statistics of all the variables in the model, together with the data sources and the reference years.[26]

4.2 | Empirical results

Once run all regressions, Table 2 collects the results of all regressions presented in Equation (7). Columns (1) and (2) display the results of the baseline and the augmented regressions related to the Creative *filière* (first row of the taxonomy); columns (3) and (4) refer to the second line dealing with Vast *filière* CCIs; columns (5) to (8) refer to the Short *filière*: the first two considering manufacturing while the last two services.

Starting with the Creative *filière*, the coefficient of the sectoral relatedness is negatively correlated with concentration of CCIs while the coefficient of the cognitive relatedness is positive and significant, instead. Augmenting the regression and considering the inventiveness as a discriminant factor, the sectoral relatedness is unexpectedly negatively correlated with both *Inventive* and *Replicative* CCIs with this *filière*. This result can be read as a congestion/competition effect due to actors with common background. A related employment base is a deterrent for the settlement of other CCIs. Contrary, cognitive relatedness follows theoretical expectations. It represents a powerful driver

[26]Variables obtained through the patents in Orbis refer to 2016 as they are accumulated values.

TABLE 1 Descriptive statistics.

Variable	Obs.	Mean	Std. Dev.	Min	Max	Source	Year
Share of regional employment in sector i (dep.var)	49,352	0.119	0.456	0	34.821	ORBIS	2015
Inventive dummy	49,352	0.494	0.500	0	1	ORBIS	2016
Sectoral relatedness (RV empl.)	49,352	5.081	2.380	0	12.966	ORBIS	2013
Cognitive relatedness (RV pat.)	49,352	0.715	1.018	0	7.109	ORBIS	2016
Sectoral unrelatedness (UV empl.)	49,352	11.084	3.177	0	17.343	ORBIS	2013
Cognitive unrelatedness (UV pat.)	49,352	2.820	2.110	0	12.794	ORBIS	2016
Labour pooling	49,352	0.402	0.651	0	6.857	ORBIS	2013–2015
Creative knowledge accumulated	49,352	0.271	2.097	0	100	ORBIS	2016
Market size (GVA)	49,352	0.125	0.203	0.002	1.813	Eurostat	2013
Technology in retail (Online Sales)	49,352	0.875	0.559	0.030	2.254	Eurostat	2008–2016
Population density (ln)	49,352	5.199	1.404	0.693	9.958	Eurostat	2013
Capital city region	49,352	0.099	0.299	0	1	Eurostat	-
UNESCO World Heritage sites	49,352	0.360	0.659	0	4	UNESCO	-
Population (ln)	49,352	12.762	0.845	9.274	15.680	Eurostat	2013
New EU Member (from 2004)	49,352	0.187	0.390	0	1	Eurostat	-
GDP (ln)	49,352	8.879	1.103	5.176	12.233	Eurostat	2013
GDP per capita (ln)	49,352	9.955	0.687	7.972	12.822	Eurostat	2013
Education levels	49,352	0.268	0.092	0.114	0.682	Eurostat	2013–2015

of industrial concentration only of innovative CCIs within this *filière*. This result is strong and coherent with theoretical priors. Thus, CCIs interacting mostly with other creative businesses tend to concentrate in space to take advantage from a cognitively related environment. Moving towards CCIs with a vast *filière*, sectoral unrelatedness shows significant coefficients on concentration but with a negative sign, both in the baseline and in the augmented regression framework. No major differences emerge between *Replicative* and *Inventive* activities. An environment characterized by a diverse creative workforce is not a factor of attraction for CCIs. Instead, cognitive unrelatedness proves its potential. Although in the baseline model it is not correlated with concentration, splitting activities according to their inventiveness allows us to detect a benefit for the *Inventive* ones, increasing spatial clustering due to a diverse creative cognitive environment. Even if this result is specular to the one of cognitive relatedness, it is theoretically more influential. Unrelatedness was usually considered as a limit to innovation and productivity (Aarstad et al., 2016) and not correlated to employment growth but rather a protection against unemployment rise (Frenken et al., 2007). In the field of CCIs, instead, the diversity of the cognitive environment is beneficial for those who innovate if included in a dynamic B2B context (Vast *filière*). The rationale is that innovative CCIs tend to concentrate in places where they can build on heterogeneous knowledge, also produced in other creative domains, thanks to the vast trade relationships they embrace in the economy. Finally, columns (6) and (7) present the results for manufacturing CCIs with a Short *filière*. Labour pooling is marginally more beneficial for the clustering of *Replicative* manufacturing activities, underlining its role as a static benefit (Combes & Duranton, 2006; Rosenthal & Strange, 2001). Instead, the role of the creative knowledge on spatial concentration is unexpected. Due to the dynamics of cumulation of knowledge in space (Balland et al., 2020; Capello, 2017), the expectation was to witness this pattern also for CCIs. Instead, a well-established stock of sectoral knowledge in a place is beneficial for all actors, both those who *replicate* it with a phenomenon of adoption and those who *invent* due to cumulation of existing knowledge, as suggested by the absence of significance of the coefficient of the interaction. Then, with services activities with a short *filière*, the

TABLE 2 Empirical econometric results.

Dep Var. Employment share in CCI *i* within region *r*	(1) Creative *filière*	(2) Creative *filière*	(3) Vast *filière*	(4) Vast *filière*	(5) Short *filière*—Man	(6) Short *filière*—Man	(7) Short *filière*—Ser	(8) Short *filière*—Ser
Inventive dummy		0.0063		−0.0883**		0.0615***		0.0571***
		(0.018)		(0.042)		(0.013)		(0.017)
Sectoral relatedness	−0.0041***	−0.0057**						
	(0.001)	(0.002)						
Cognitive relatedness	0.0046*	−0.0215***						
	(0.002)	(0.005)						
Inventive dummy × *Sectoral relatedness*		0.0022						
		(0.003)						
Inventive dummy × *Cognitive relatedness*		0.0293***						
		(0.005)						
Sectoral unrelatedness			−0.0108***	−0.0130***				
			(0.003)	(0.004)				
Cognitive unrelatedness			0.0075**	−0.0064†				
			(0.004)	(0.004)				
Inventive dummy × *Sectoral unrelatedness*				0.0041				
				(0.004)				
Inventive dummy × *Cognitive unrelatedness*				0.0208***				
				(0.004)				
Labour Pooling					0.0219***	0.0337***		
					(0.008)	(0.01)		
Creative Knowledge accumul.					0.0369***	0.0406***		
					(0.009)	(0.016)		
Inventive dummy × *Labour Pooling*						−0.0469***		
						(0.015)		

TABLE 2 (Continued)

Dep Var. Employment share in CCI *i* within region *r*	(1) Creative *filière*	(2) Creative *filière*	(3) Vast *filière*	(4) Vast *filière*	(5) Short *filière*—Man	(6) Short *filière*—Man	(7) Short *filière*—Ser	(8) Short *filière*—Ser
Inventive dummy × Creative Knowledge accumul.						−0.0071		
						(0.019)		
Mkt. size							0.0505*	0.1425***
							(0.029)	(0.047)
Tech. in retail							0.054	0.028
							(0.042)	(0.044)
Inventive dummy × Mkt. size								−0.1082***
								(0.038)
Inventive dummy × Tech. in retail								0.0433**
								(0.018)
Population density (ln)	0.0023	0.0027	0.0133**	0.0133**	0.0134**	0.0138***	0.0163**	0.0159**
	(0.003)	(0.003)	(0.006)	(0.006)	(0.005)	(0.005)	(0.007)	(0.007)
Capital city region	0.0298**	0.0294**	0.022	0.0238	−0.0539***	−0.0498***	0.0164	0.0195
	(0.012)	(0.012)	(0.018)	(0.018)	(0.019)	(0.018)	(0.02)	(0.02)
UNESCO World Heritage sites	0.0051	0.0049	−0.0021	−0.0021	0.0115	0.0118	0.0045	0.0043
	(0.004)	(0.004)	(0.008)	(0.008)	(0.01)	(0.01)	(0.005)	(0.005)
Population (ln)	−0.0135**	−0.0140***	0.0147	0.0155	−0.0422***	−0.0471***	−0.0243***	−0.0320***
	(0.005)	(0.005)	(0.014)	(0.014)	(0.007)	(0.007)	(0.009)	(0.009)
New EU Member (from 2004)	0.0081	0.012	0.3071***	0.3076***	0.0570*	0.0569*	0.0668	0.0668
	(0.025)	(0.025)	(0.045)	(0.046)	(0.032)	(0.031)	(0.049)	(0.048)
GDP per capita (ln)	0.0259***	0.0262***	0.1308***	0.1328***	−0.0469***	−0.0520***	0.0415**	0.0354**
	(0.007)	(0.007)	(0.022)	(0.022)	(0.014)	(0.014)	(0.017)	(0.017)
Population 25–64 with tertiary education (%)	0.0765	0.0714	0.3947***	0.3809***	0.12	0.1063	0.2514†	0.2172
	(0.061)	(0.06)	(0.082)	(0.083)	(0.111)	(0.11)	(0.169)	(0.169)

(Continues)

TABLE 2 (Continued)

Dep Var. Employment share in CCI *i* within region *r*	(1) Creative *filière*	(2) Creative *filière*	(3) Vast *filière*	(4) Vast *filière*	(5) Short *filière*—Man	(6) Short *filière*—Man	(7) Short *filière*—Ser	(8) Short *filière*—Ser
Constant	0.0711	0.0846	−1.5426***	−1.5130***	1.0827***	1.1837***	−0.2062	−0.061
	(0.104)	(0.104)	(0.302)	(0.307)	(0.154)	(0.155)	(0.227)	(0.228)
Observations	7818	7818	7872	7872	10,728	10,728	22,934	22,934
Number of regions	1254	1254	1275	1275	1249	1249	1321	1321
Model	Random Effect	Random Effect	Random Effect	Random Effect	Random Effect	Random Effect	Random Effect	Random Effect
Industry FE	YES	YES	YES	YES	YES	YES	YES	YES
Country FE	YES	YES	YES	YES	YES	YES	YES	YES
R2overall	0.138	0.144	0.18	0.185	0.0901	0.0921	0.0977	0.099

Notes: Robust standard errors in parentheses.
$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$, $^{\dagger}p < 0.15$.

focus shifts towards retail and cultural services. As expected, the market potential is beneficial for service *Replicative* activities, the coefficient of GVA is positive and significant only for them, confirming its static role. On the dynamic side, a higher propensity of people to technology in retail (measured through the online sales) is associated with a higher spatial concentration of market-oriented services activities that are also *Inventive*, meaning that expert and demanding consumers stimulate the development of new innovations.

To conclude the analysis of the results, comments on the external validity are worthwhile. The EU being the geographical scope of the analysis, the results are not necessarily valid in other contexts. Even if at different scales, the EU represents an island of innovation and property right protection compared with other areas such as developing countries. Thus, the logics of clustering may be different in other settings. In developing countries, for instance, multi-national enterprises drive the innovativeness of areas, perhaps changing the results (Anand et al., 2021). However, the methodology remains valid and replicable. In fact, also in areas with low IPRs intensity, it would be possible to distinguish between those that are *Inventive* from those that are *Replicative*.

4.3 | Robustness checks

In order to further validate the results, some robustness checks were performed. First, the same models were re-estimated without subdividing CCIs according to their *filière* to discuss the role of the creative process in better clarifying the reasons behind CCIs' spatial clustering (Table A3 in the Appendix). The check reinforced the interpretation on the role of *Inventive*ness in distinguishing the needs of CCIs at the territorial level; otherwise, its relevance would be downsized. Without the subdivision into *filières*, in fact, the *Inventive* dummy would not be able to distinguish between static and dynamic benefits, especially for non-market-oriented CCIs, highlighting the importance of the cognitive knowledge mix present at the local level as dynamic benefit. Thanks to trade exchanges, industries generate value and innovations. In Table A3, the columns' headings refer to *filières* to ease comparison. Considering *Creative* and *Vast filières*, the interaction of cognitive (un)relatedness with the *Inventive* dummy is not statistically significant, failing in detecting the innovation stimulus for CCIs. This confirms the importance of separating B2B from market-oriented CCIs: cognitive measures reflect dynamic agglomeration economies only if industries participate actively in the *filière*, trading goods and services producing value. This *learning by exchanging* process enhances the mechanism of creativity generation in CCIs. Market-oriented CCIs, instead, innovate more randomly and local knowledge attract also non-innovators that benefit from a cognitively lively environment. In fact, a knowledge-rich environment reflects the reputation and the position of actors operating in it that will have an advantage in facing the market.

Second, embedded in the taxonomy there is the idea that the agglomerative drivers for CCIs' clustering are *filière*-specific (e.g., cognitive relatedness works for creative *filière* CCIs and not for short *filière* ones). However, overlapping of agglomeration forces can occur, suggesting to re-estimate the model including at the same time all the variables included in the taxonomy. However, the inclusion of additional variables may have a twofold effect: on the one hand, the fit could improve; on the other hand, potential multicollinearity could generate statistical bias leading to misinterpretation of the results.

Thus, as a further check, every specification has been re-estimated including all explanatory variables, also those related to other rows of the taxonomy (Table A4 in the Appendix). Results show minor but relevant changes, especially for the *short filière*. The coefficient of creative knowledge has the largest size in this segment, corroborating the idea that accumulated knowledge strongly matters for these activities. Furthermore, cognitive measures of relatedness are negligible for them, putting the accent on the importance of trade. In fact, if activities trade they are interconnected with either similar or different economic actors (creative or not) and the cognitive dimension matters for better exchanges. Therefore, the taxonomy helps in better detecting the determinants of spatial clustering.

5 | CONCLUSIONS AND IMPLICATIONS

This work emphasizes that creativity in CCIs is a multifaceted aspect. Not all CCIs express the same form and intensity of creativity and this feature is reflected by their spatial pattern. This means that the location determinants behind spatial concentration differ based on the forms and intensity of the creativity expressed. Besides the heterogeneous innovative behaviour, CCIs' also differ for their various trade relationships (*filière*) with the rest of the economy.

Empirical results confirm that the heterogeneous creativity of CCIs explains different localization drivers only if *filière* effects are considered. Therefore, the location determinants are diversified and, despite their similarities, CCIs tend to display different spatial patterns according to the heterogeneity expressed. This result has important conceptual implications, stressing the importance of not considering CCIs as a unique object, but classifying them in a coherent way.

Although this work builds on an industrial approach to the creative economy, connections with the creative class theory are as relevant as ever. Compared to the industrial approach, focused on how creativity turns into innovation and economic value, in the creative class theory (Florida, 2002) this is reflected into what people do, that is, their occupations. Assuming that *Inventive* and *Replicative* CCIs tend to cluster in different places based on the presence of different territorial conditions, as found in this work, the typology of creative workers will also be different. Within *Inventive* CCIs, there will be more occupations related to the conception and invention phases; while, within *Replicative* ones, there will be more occupations related to production and distribution, with strong implications for growth and development. This could be a starting point for future studies.

Findings of these paper are also relevant for the implications of culture and creativity to local economic development through different channels such as cognitive aspects (Berti Mecocci et al., 2022) and cultural participation (Cerisola & Panzera, 2022). Agglomeration of CCIs is important because CCIs need to reach a certain threshold size in order to trigger positive spillovers in the local economy (Cicerone et al., 2021).

Building on these results, policy-makers aimed at attracting CCIs should consider that not all CCIs respond to the same stimuluses as their creativity is not homogeneous and that their *filières* may be variegated. Local authorities should elaborate place-based strategies for the development of CCIs, considering the peculiar innovative behaviour that they put in place and the specificities of the local areas. Moreover, local authorities may promote targeted policies for different *filières* of CCIs, to reinforce the multiplicative effects of territorial conditions. Finally, as different innovations drive different localization behaviours, the innovative behaviour of CCIs should be considered as stimulus to economic growth.

This study presents, however, some limitations, summed up into two points. First, this work has no time variation. It does not analyse the dynamics of concentration over time, mostly due to data availability. Future research on this theme should focus on the dynamics of spatial clustering over time. Second, although the theoretical framework poses the attention also on the different ways in which creativity expresses, presenting the three forms of innovation related to it (patents, trademarks and copyrights), this empirical analysis only disentangled the different territorial conditions that foster the spatial concentration comparing *Replicative* and *Inventive* activities. Future studies need to deepen the analysis focusing on the creative modes within *Inventive* activities, trying to clarify the local drivers of concentration of different creative actors.

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APPENDIX

TABLE A1 List of sectors belonging to CCIs according to Santagata's classification and descriptive stats of empl. share.

NACE Rev.2	NACE Rev.2 Label	Copyright intense	Statistics of empl. share for each industry: Mean	St. Dev.	Min.	Max.	Skewness
1013	Production of meat and poultry meat products		0.17%	0.28%	0.00%	2.56%	3.98
1051	Operation of dairies and cheese making		0.15%	0.22%	0.00%	2.02%	3.73
1102	Manufacture of wine from grape		0.10%	0.18%	0.00%	2.12%	5.46
1310	Preparation and spinning of textile fibres		0.13%	0.48%	0.00%	6.09%	9.53
1320	Weaving of textiles		0.12%	0.33%	0.00%	3.32%	6.21
1330	Finishing of textiles		0.06%	0.21%	0.00%	2.97%	9.42
1391	Manufacture of knitted and crocheted fabrics		0.05%	0.10%	0.00%	0.77%	4.13
1393	Manufacture of carpets and rugs		0.08%	0.26%	0.00%	2.74%	7.72
1399	Manufacture of other textiles n.e.c.		0.05%	0.10%	0.00%	1.05%	5.45
1411	Manufacture of leather clothes		0.02%	0.05%	0.00%	0.34%	4.74
1413	Manufacture of other outerwear		0.44%	1.74%	0.00%	34.82%	15.02
1414	Manufacture of wearing apparel		0.18%	0.43%	0.00%	4.42%	6.14
1419	Manufacture of other wearing apparel and accessories		0.08%	0.22%	0.00%	2.59%	7.18
1420	Manufacture of articles of fur		0.05%	0.37%	0.00%	3.76%	9.89
1431	Manufacture of knitted and crocheted hosiery		0.10%	0.32%	0.00%	2.89%	5.98
1439	Manufacture of other knitted and crocheted apparel		0.09%	0.18%	0.00%	1.30%	4.00
1511	Tanning and dressing of leather; dressing and dyeing of fur		0.12%	0.49%	0.00%	5.85%	8.66
1512	Manufacture of luggage, handbags and the like, saddlery and harness		0.08%	0.19%	0.00%	2.02%	5.80
1520	Manufacture of footwear		0.33%	0.86%	0.00%	10.70%	6.55
1623	Manufacture of other builders' carpentry and joinery		0.14%	0.22%	0.00%	2.94%	5.04
1629	Manufacture of other products of wood; manufacture of articles of cork, straw and plaiting materials		0.07%	0.13%	0.00%	1.22%	4.39
1811	Printing of newspapers	x	0.06%	0.14%	0.00%	1.73%	7.49
1812	Other printing	x	0.14%	0.15%	0.00%	1.92%	3.47
1813	Pre-press and pre-media services	x	0.03%	0.13%	0.00%	2.98%	19.03
1814	Binding and related services	x	0.02%	0.05%	0.00%	0.78%	9.13
1820	Reproduction of recorded media	x	0.02%	0.08%	0.00%	0.92%	8.12
2319	Manufacture and processing of other glass, including technical glassware		0.10%	0.32%	0.00%	3.74%	7.42

(Continues)

TABLE A1 (Continued)

NACE Rev.2	NACE Rev.2 Label	Copyright intense	Statistics of empl. share for each industry: Mean	St. Dev.	Min.	Max.	Skewness
2331	Manufacture of ceramic tiles and flags		0.14%	0.49%	0.00%	4.09%	5.58
2341	Manufacture of ceramic household and ornamental articles		0.08%	0.25%	0.00%	2.24%	5.64
2370	Cutting, shaping and finishing of stone		0.06%	0.10%	0.00%	1.53%	6.74
2512	Manufacture of doors and windows of metal		0.09%	0.11%	0.00%	0.79%	2.79
2571	Manufacture of cutlery		0.06%	0.24%	0.00%	2.50%	9.37
2630	Manufacture of communication equipment		0.08%	0.17%	0.00%	1.44%	4.21
2640	Manufacture of consumer electronics		0.08%	0.43%	0.00%	6.66%	13.63
2652	Manufacture of watches and clocks		0.04%	0.14%	0.00%	1.31%	7.43
2740	Manufacture of electric lighting equipment		0.08%	0.16%	0.00%	1.85%	5.39
3101	Manufacture of office and shop furniture		0.08%	0.14%	0.00%	1.60%	4.88
3102	Manufacture of kitchen furniture		0.07%	0.17%	0.00%	2.28%	8.10
3109	Manufacture of other furniture		0.22%	0.45%	0.00%	4.99%	5.15
3212	Manufacture of jewellery and related articles		0.04%	0.20%	0.00%	3.00%	11.13
3220	Manufacture of musical instruments		0.03%	0.06%	0.00%	0.50%	5.10
3240	Manufacture of games and toys		0.07%	0.25%	0.00%	3.96%	11.54
4616	Agents involved in the sale of textiles, clothing, fur, footwear and leather goods		0.02%	0.04%	0.00%	0.39%	5.00
4624	Wholesale of hides, skins and leather		0.01%	0.03%	0.00%	0.40%	7.13
4641	Wholesale of textiles		0.04%	0.09%	0.00%	0.90%	5.57
4642	Wholesale of clothing and footwear		0.08%	0.14%	0.00%	1.78%	5.27
4643	Wholesale of electrical household appliances		0.08%	0.13%	0.00%	2.11%	6.67
4647	Wholesale of furniture, carpets and lighting equipment		0.04%	0.09%	0.00%	1.44%	8.91
4649	Wholesale of other household goods		0.12%	0.19%	0.00%	3.18%	7.08
4651	Wholesale of computers, computer peripheral equipment and software		0.07%	0.13%	0.00%	1.41%	4.86
4719	Other retail sale in non-specialized stores		0.25%	1.16%	0.00%	24.79%	14.81
4722	Retail sale of meat and meat products in specialized stores		0.06%	0.14%	0.00%	3.07%	15.47
4725	Retail sale of beverages in specialized stores		0.03%	0.09%	0.00%	1.42%	10.57
4729	Other retail sale of food in specialized stores		0.05%	0.11%	0.00%	1.51%	6.69

TABLE A1 (Continued)

NACE Rev.2	NACE Rev.2 Label	Copyright intense	Statistics of empl. share for each industry: Mean	St. Dev.	Min.	Max.	Skewness
4743	Retail sale of audio and video equipment in specialized stores		0.05%	0.25%	0.00%	6.23%	22.25
4751	Retail sale of textiles in specialized stores		0.05%	0.16%	0.00%	2.65%	11.39
4759	Retail sale of furniture, lighting equipment and other household articles in specialized stores		0.16%	0.70%	0.00%	16.00%	19.17
4761	Retail sale of books in specialized stores	x	0.02%	0.03%	0.00%	0.48%	8.09
4762	Retail sale of newspapers and stationery in specialized stores	x	0.04%	0.10%	0.00%	1.16%	8.11
4763	Retail sale of music and video recordings in specialized stores	x	0.01%	0.03%	0.00%	0.34%	10.24
4771	Retail sale of clothing in specialized stores		0.28%	0.96%	0.00%	15.11%	10.36
4777	Retail sale of watches and jewellery in specialized stores		0.04%	0.10%	0.00%	1.86%	12.81
5510	Hotels and similar accommodation		0.46%	0.88%	0.00%	11.99%	5.90
5520	Holiday and other short-stay accommodation		0.06%	0.11%	0.00%	1.03%	4.41
5610	Restaurants and mobile food service activities		0.59%	1.37%	0.00%	27.93%	14.83
5811	Book publishing	x	0.04%	0.09%	0.00%	1.25%	7.02
5813	Publishing of newspapers	x	0.07%	0.16%	0.00%	1.85%	6.87
5814	Publishing of journals and periodicals	x	0.05%	0.15%	0.00%	3.07%	15.53
5821	Publishing of computer games	x	0.01%	0.02%	0.00%	0.11%	3.32
5829	Other software publishing		0.05%	0.11%	0.00%	1.16%	5.24
5911	Motion picture, video and television programme production activities	x	0.05%	0.22%	0.00%	4.51%	16.51
5912	Motion picture, video and television programme post-production activities	x	0.02%	0.04%	0.00%	0.42%	6.98
5913	Motion picture, video and television programme distribution activities	x	0.02%	0.15%	0.00%	1.90%	11.77
5914	Motion picture projection activities	x	0.03%	0.11%	0.00%	1.71%	10.92
5920	Sound recording and music publishing activities	x	0.01%	0.02%	0.00%	0.18%	5.43
6010	Radio broadcasting	x	0.02%	0.03%	0.00%	0.29%	3.99
6020	Television programming and broadcasting activities	x	0.04%	0.11%	0.00%	0.86%	4.98
6201	Computer programming activities	x	0.23%	0.37%	0.00%	4.77%	5.64
6202	Computer consultancy activities	x	0.14%	0.35%	0.00%	5.27%	9.28

(Continues)

TABLE A1 (Continued)

NACE Rev.2	NACE Rev.2 Label	Copyright intense	Statistics of empl. share for each industry Mean	St. Dev.	Min.	Max.	Skewness
6209	Other information technology and computer service activities	x	0.13%	0.29%	0.00%	4.27%	7.17
6391	News agency activities	x	0.01%	0.04%	0.00%	0.44%	7.12
7111	Architectural activities		0.06%	0.09%	0.00%	1.18%	6.36
7112	Engineering activities and related technical consultancy		0.36%	0.48%	0.00%	7.95%	6.06
7311	Advertising agencies	x	0.10%	0.18%	0.00%	2.40%	5.59
7312	Media representation	x	0.02%	0.04%	0.00%	0.52%	5.88
7410	Specialized design activities	x	0.03%	0.08%	0.00%	1.72%	16.90
9001	Performing arts	x	0.03%	0.05%	0.00%	0.62%	5.38
9002	Support activities to performing arts	x	0.02%	0.05%	0.00%	0.82%	11.45
9003	Art Creation	x	0.01%	0.04%	0.00%	0.62%	12.08
9004	Operation of arts facilities	x	0.05%	0.12%	0.00%	1.31%	5.80
9101	Library and archives activities	x	0.02%	0.04%	0.00%	0.36%	5.40
9102	Museums activities		0.02%	0.06%	0.00%	0.65%	7.30
9103	Operation of historical sites and buildings and similar visitor attractions		0.03%	0.25%	0.00%	3.71%	14.45
9104	Botanical and zoological gardens and nature reserves activities		0.03%	0.05%	0.00%	0.47%	5.05
9329	Other amusement and recreation activities	x	0.07%	0.11%	0.00%	0.99%	4.55

TABLE A2 Classification of CCIs according to IO trade relationships (in italic manufacturing).

Industry	Final demand flow (% of total flow)	Intermediate flow with CCIs (% of intermediate flow)	Intermediate flow with non-CCIs industries (% of intermediate flow)	*Filière*
Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials (C16)	21.1%	45.0%	55.0%	Creative *filière*
Printing and reproduction of recorded media (C18)	13.6%	51.7%	48.3%	Creative *filière*
Motion picture, video, television programme production; programming and broadcasting activities (J59_60)	45.2%	56.2%	43.8%	Creative *filière*
Advertising and market research (M73)	3.7%	50.6%	49.4%	Creative *filière*
Manufacture of other non-metallic mineral products (C23)	18.7%	27.4%	72.6%	Vast *filière*
Manufacture of fabricated metal products, except machinery and equipment (C25)	25.6%	33.1%	66.9%	Vast *filière*
Computer programming, consultancy and information service activities (J62_63)	45.4%	35.8%	64.2%	Vast *filière*
Architectural and engineering activities; technical testing and analysis (M71)	25.6%	30.2%	69.8%	Vast *filière*
Other professional, scientific and technical activities; veterinary activities (M74_75)	31.4%	37.2%	62.8%	Vast *filière*
Manufacture of food products; beverages and tobacco products (C10T12)	63.5%	73.8%	26.2%	Short *filière*
Manufacture of textiles, wearing apparel, leather and related products (C13T15)	64.5%	67.0%	33.0%	Short *filière*
Manufacture of computer, electronic and optical products (C26)	65.4%	42.6%	57.4%	Short *filière*
Manufacture of electrical equipment (C27)	50.9%	35.7%	64.3%	Short *filière*
Manufacture of furniture; other manufacturing (C31_32)	72.2%	34.6%	65.4%	Short *filière*
Wholesale trade, except of motor vehicles and motorcycles (G46)	50.4%	40.0%	60.0%	Short *filière*
Retail trade, except of motor vehicles and motorcycles (G47)	72.1%	35.9%	64.1%	Short *filière*
Accommodation and food service activities (I)	80.5%	29.0%	71.0%	Short *filière*
Publishing activities (J58)	51.1%	46.0%	54.0%	Short *filière*
Creative, arts and entertainment activities; libraries, archives, museums and other cultural activities; gambling and betting activities (R90T92)	77.9%	62.4%	37.6%	Short *filière*
Sports activities and amusement and recreation activities (R93)	68.5%	56.2%	43.8%	Short *filière*
Mean of all CCIs	**47.4%**	**44.5%**	**55.5%**	

TABLE A3 Empirical results without separating CCIs into *filières*.

Dep. Var. Employment share in CCI *i* within the region *r*	(1) Creative *filière*	(2) Creative *filière*	(3) Vast *filière*	(4) Vast *filière*	(5) Short *filière*—Man	(6) Short *filière*—Man	(7) Short *filière*—Ser	(8) Short *filière*—Ser
Inventive dummy		0.0505**		0.0337		0.0452***		0.0333**
		(0.022)		(0.042)		(0.011)		(0.014)
Sectoral relatedness	−0.0090***	−0.0082***						
	(0.002)	(0.003)						
Cognitive relatedness	0.0138***	0.0104*						
	(0.003)	(0.006)						
Inventive dummy × *Sectoral relatedness*		−0.0017						
		(0.004)						
Inventive dummy × *Cognitive relatedness*		0.0040						
		(0.007)						
Sectoral unrelatedness			−0.0162***	−0.0168***				
			(0.002)	(0.003)				
Cognitive unrelatedness			0.0140***	0.0126***				
			(0.002)	(0.004)				
Inventive dummy × *Sectoral unrelatedness*				0.0010				
				(0.004)				
Inventive dummy × *Cognitive unrelatedness*				0.0003				
				(0.004)				
Labour Pooling					0.0144***	0.0238***		
					(0.005)	(0.007)		
Creative Knowledge accumul.					0.0357***	0.0394***		
					(0.007)	(0.013)		
Inventive dummy × *Labour Pooling*						−0.0281***		
						(0.009)		

TABLE A3 (Continued)

Dep. Var. Employment share in CCI *i* within the region *r*	(1) Creative *filière*	(2) Creative *filière*	(3) Vast *filière*	(4) Vast *filière*	(5) Short *filière*—Man	(6) Short *filière*—Man	(7) Short *filière*—Ser	(8) Short *filière*—Ser
Inventive dummy × Creative Knowledge accumul.						−0.0067		
						(0.015)		
Mkt. size							0.0737***	0.0955**
							(0.027)	(0.042)
Inventive dummy × Mkt. size								−0.0221
								(0.034)
Tech. in retail							0.0405	0.0141
							(0.031)	(0.033)
Inventive dummy × Tech. in retail								0.0406***
								(0.014)
Population density (ln)	0.0148***	0.0148***	0.0146***	0.0145***	0.0065*	0.0067*	0.0151***	0.0149***
	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.004)	(0.005)	(0.005)
Capital city region	0.0094	0.0114	0.0111	0.0127	−0.0330***	−0.0305**	0.0190	0.0214†
	(0.011)	(0.011)	(0.011)	(0.010)	(0.013)	(0.013)	(0.015)	(0.015)
UNESCO World Heritage sites	0.0034	0.0036	0.0023	0.0023	0.0075	0.0077	0.0039	0.0040
	(0.004)	(0.004)	(0.004)	(0.004)	(0.007)	(0.007)	(0.005)	(0.005)
Population (ln)	−0.0220***	−0.0245***	−0.0106†	−0.0115†	−0.0341***	−0.0371***	−0.0189***	−0.0224***
	(0.005)	(0.005)	(0.007)	(0.007)	(0.005)	(0.005)	(0.007)	(0.007)
New EU Member (from 2004)	0.0771***	0.0815***	0.1369***	0.1421***	0.0295	0.0307	0.1193***	0.1192***
	(0.021)	(0.021)	(0.021)	(0.021)	(0.024)	(0.024)	(0.038)	(0.038)
GDP per capita (ln)	0.0277***	0.0257**	0.0333***	0.0324***	−0.0268***	−0.0296***	0.0592***	0.0571***
	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)	(0.010)	(0.013)	(0.013)
Population 25–64 with tertiary education (%)	0.1480**	0.1371*	0.1398**	0.1332*	0.0711	0.0636	0.2707**	0.2518**
	(0.074)	(0.074)	(0.068)	(0.068)	(0.077)	(0.077)	(0.125)	(0.126)

(Continues)

TABLE A3 (Continued)

Dep. Var. Employment share in CCI *i* within the region *r*	(1) Creative *filière*	(2) Creative *filière*	(3) Vast *filière*	(4) Vast *filière*	(5) Short *filière*—Man	(6) Short *filière*—Man	(7) Short *filière*—Ser	(8) Short *filière*—Ser
Constant	0.1259	0.1651	0.0235	0.0425	0.8290***	0.8848***	−0.4846***	−0.4245**
	(0.124)	(0.124)	(0.139)	(0.139)	(0.113)	(0.113)	(0.179)	(0.180)
Observations	49,352	49,352	49,352	49,352	17,021	17,021	32,331	32,331
Number of regions	1327	1327	1327	1327	1304	1304	1323	1323
Model	Random Effect	Random Effect	Random Effect	Random Effect	Random Effect	Random Effect	Random Effect	Random Effect
Industry FE	YES	YES	YES	YES	YES	YES	YES	YES
Country FE	YES	YES	YES	YES	YES	YES	YES	YES
R2 overall	0.0875	0.0886	0.0898	0.0910	0.0919	0.0934	0.107	0.108

Notes: Robust standard errors in parentheses. *Filières* indicated in the columns make reference to the main specification to simplify comparisons.
$^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$, $^{\dagger}p < 0.15$.

TABLE A4 Results including all localization factors.

Dep Var. Share of employment in CCI *i* within the region *r*	**(1) Creative *filière***	**(2) Vast *filière***	**(3) Short *filière*—Man**	**(4) Short *filière*—Ser**
Inventive dummy	−0.0113	−0.1299***	0.0609***	0.0794***
	(0.018)	(0.042)	(0.016)	(0.020)
Sectoral relatedness	−0.0075***	−0.0081**	−0.0071	0.0145***
	(0.003)	(0.003)	(0.006)	(0.003)
Cognitive relatedness	−0.0319***	−0.0022	0.0119	−0.0136*
	(0.006)	(0.006)	(0.009)	(0.007)
Sectoral unrelatedness	0.0040**	−0.0071†	−0.0114***	−0.0328***
	(0.002)	(0.005)	(0.004)	(0.006)
Cognitive unrelatedness	0.0101***	−0.0076†	0.0051	0.0240***
	(0.002)	(0.005)	(0.008)	(0.006)
Labour Pooling	0.0040*	0.0036	0.0348***	
	(0.002)	(0.010)	(0.009)	
Creative Knowledge accumul.	0.0169***	0.0122***	0.0399**	
	(0.006)	(0.003)	(0.016)	
Mkt. size	0.0590***	0.0535		0.0310
	(0.023)	(0.061)		(0.055)
Tech. in retail	0.0081	0.0384		0.0353
	(0.021)	(0.039)		(0.041)
Inventive dummy × *Sectoral relatedness*	0.0024			
	(0.003)			
Inventive dummy × *Cognitive relatedness*	0.0290***			
	(0.005)			
Inventive dummy × *Sectoral unrelatedness*		0.0050		
		(0.004)		
Inventive dummy × *Cognitive unrelatedness*		0.0228***		
		(0.004)		
Inventive dummy × *Labour Pooling*			−0.0490***	
			(0.015)	
Inventive dummy × *Creative Knowledge accumul.*			−0.0066	
			(0.019)	
Inventive dummy × *Mkt. size*				−0.1007***
				(0.037)
Inventive dummy × *Tech. in retail*				−0.0207
				(0.019)
Population density (ln)	0.0039†	0.0132**	0.0152***	0.0161**
	(0.003)	(0.006)	(0.005)	(0.007)
Capital city region	0.0289**	0.0240	−0.0470**	0.0281†
	(0.012)	(0.018)	(0.019)	(0.019)

(Continues)

TABLE A4 (Continued)

Dep Var. Share of employment in CCI *i* within the region *r*	(1) Creative *filière*	(2) Vast *filière*	(3) Short *filière*—Man	(4) Short *filière*—Ser
UNESCO World Heritage sites	0.0027	−0.0051	0.0061	0.0051
	(0.004)	(0.008)	(0.010)	(0.005)
Population (ln)	−0.0371***	0.0087	−0.0307*	0.0042
	(0.006)	(0.016)	(0.018)	(0.013)
New EU Member (from 2004)	0.0108	0.3172***	0.1113***	0.1863***
	(0.029)	(0.057)	(0.033)	(0.051)
GDP per capita (ln)	0.0059	0.1220***	−0.0482***	0.0580***
	(0.007)	(0.020)	(0.016)	(0.016)
Population 25–64 with tertiary education (%)	0.0173	0.3153***	0.1023	0.0864
	(0.069)	(0.094)	(0.112)	(0.137)
Constant	0.5110***	−1.3711***	1.0737***	−0.5053**
	(0.124)	(0.304)	(0.306)	(0.238)
Observations	7818	7872	10,728	22,934
Number of regions	1254	1275	1249	1321
Model	Random Effect	Random Effect	Random Effect	Random Effect
Industry FE	YES	YES	YES	YES
Country FE	YES	YES	YES	YES
R2 overall	0.165	0.205	0.0957	0.105

Notes: Robust standard errors in parentheses.
**p* < 0.1, ***p* < 0.05, ****p* < 0.01, †*p* < 0.15.

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Resumen. La concentración espacial de las Industrias Culturales y Creativas (ICC) no es un tema novedoso en la investigación académica. Sin embargo, el análisis de este fenómeno suele pasar por alto que las ICC se comportan de forma diferente, debido a su heterogeneidad. Partiendo de la literatura sobre la concentración de las ICC, este artículo presenta una clasificación novedosa de las ICC basada en dos dimensiones clave: la creatividad heterogénea y la filière (sector de empresas interconectadas). La concentración espacial de las ICC se pone a prueba a lo largo de estas dos dimensiones, destacando que los factores determinantes de la localización difieren en su intersección. Los resultados renuevan el interés por la agrupación de las ICC y definen nuevas perspectivas para las intervenciones políticas.

抄録: 文化創造産業(Cultural and Creative Industries:CCI)の空間的集中は、新しい研究トピックではない。しかし、この現象の分析においては、CCIの不均一性のためにCCIの挙動が異なっているということが無視されることが多い。CCIの空間的集中に関する研究論文では、CCIの新しい分類が、異種の創造性とフィリエール(filière:生産連鎖あるは価値連鎖でつながる企業間分業)という2つの重要な側面に基づいて提示されている。CCIの空間的集中は、これら2つの次元に沿ったテストから、立地の決定要因は交差点で異なることが明らかになる。結果は、CCIのクラスタリングに関する新たな関心事を示し、政策介入における新しい視点を定義するものである。